



Portfolio & Projects

Update Documentation

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1. Portfolio Website

Repo	JerrelAdriel/JerrelAdriel.github.io
Live	https://jerreladriel.github.io
Tech Stack	HTML5, CSS3, Vanilla JavaScript
Total Sections	Hero, About, Experience, Projects, Skills, Contact

Apa yang Dibangun

Single-page portfolio dengan 6 section utama. Bilingual EN/ID dengan toggle, responsive untuk desktop & mobile, dark-mode, dengan animasi smooth. Lightbox gallery untuk preview screenshot tiap project.

Key Features

Bilingual EN/ID toggle	i18n.js — localStorage persistence, default EN
Project lightbox gallery	portfolio.js — keyboard nav, ESC close
Smooth scroll navigation	Anchor links + IntersectionObserver fade-in
Photo transparent BG	AI-processed via rembg (U ² -Net model)
Work Experience timeline	Pulsing dot animation untuk current job

Code Highlight: Bilingual System

Pattern yang dipakai: **HTML attribute-based translation**. Element ditandai dengan `data-i="key"`, lalu JavaScript replace innerHTML saat user toggle bahasa.

```
const TRANSLATIONS = {
  en: { 'about.title': 'About', /* ... */ },
  id: { 'about.title': 'Tentang', /* ... */ },
};

function applyLanguage(lang) {
  const dict = TRANSLATIONS[lang];
  document.querySelectorAll('[data-i]').forEach(el => {
    if (dict[el.dataset.i]) el.innerHTML = dict[el.dataset.i];
  });
  localStorage.setItem('portfolio_lang', lang);
}
```

■ **Note:** Default English terlihat dari HTML asli → SEO-friendly. Vanilla JS tanpa framework. Tambah element baru? Cukup tambah data-i + key di dictionary.

Code Highlight: Responsive Avatar Blur

Bug yang sempat terjadi: blur ungu di belakang foto profil tidak ikut foto saat resize. **Solusi:** pakai pseudo-element `::before` dengan padding-bottom trick untuk maintain aspect ratio square.

```
.avatar::before {
  content: '';
  position: absolute;
  width: 92%;
  padding-bottom: 92%;      /* bikin selalu square */
  top: 5%;
  left: 50%;
  transform: translateX(-50%); /* centered horizontal */
  background: radial-gradient(circle at 50% 50%,
    rgba(124,58,237,0.4), transparent 65%);
  border-radius: 50%;
  z-index: -1;              /* di belakang foto */
}
```

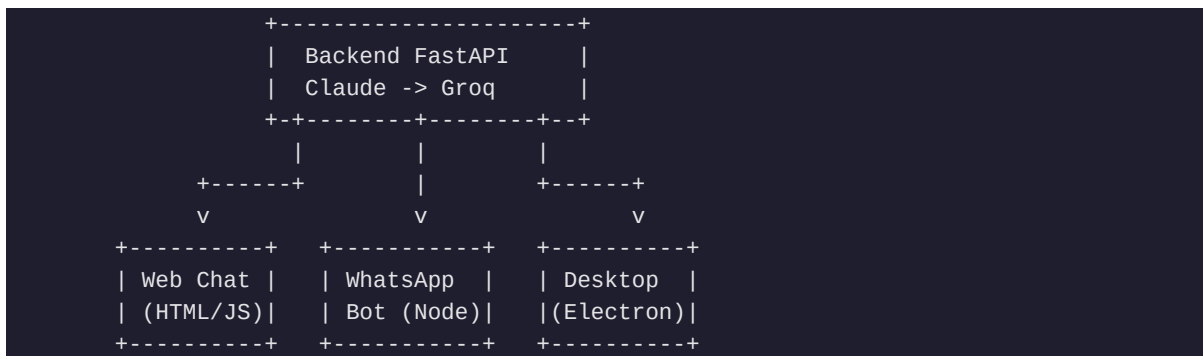
2. Agent-JR — Multi-Platform AI Assistant

Repo	JerrelAdriel/agent-jr
Tech Stack	Python FastAPI, Node.js, Electron, HTML/JS
AI Engine	Claude (Anthropic) → Groq fallback
Platforms	Web · WhatsApp Bot · Desktop App

Apa Itu?

AI assistant pribadi yang berjalan di **3 platform sekaligus** dengan satu backend FastAPI. Powered by Claude dengan fallback otomatis ke Groq Llama 3.1 kalau Claude tidak tersedia. Cocok sebagai showcase kemampuan integrasi multi-platform + AI API.

Arsitektur



Code Highlight: Fallback Claude → Groq

```
@app.post("/api/chat")
async def chat(request: ChatRequest):
    # Coba Claude dulu (lebih bagus tapi berbayar)
    if anthropic_client:
        try:
            response = anthropic_client.messages.create(
                model="claude-haiku-4-5",
                system=SYSTEM_PROMPT,
                messages=to_messages(request.messages)
            )
            return {"reply": response.content[0].text, "source": "claude"}
        except Exception as e:
            print(f"[Claude gagal] {e} -> Groq...")

    # Fallback ke Groq (gratis 14k req/hari)
    if groq_client:
        response = groq_client.chat.completions.create(
            model="llama-3.1-8b-instant",
```

```

        messages=msgsg
    )
    return {"reply": response.choices[0].message.content, "source": "groq"}

```

■ **Note:** Try-catch + fallback = graceful degradation. Field source di response biar frontend tahu API mana yang dipakai (debug).

Code Highlight: WhatsApp Mention Detection

Di grup, bot hanya merespon kalau (a) di-tag, atau (b) pesannya di-reply. **Tricky:** WhatsApp Web sekarang punya 2 format ID (@c.us legacy + @lid baru). Solusi: pakai msg.getMentions() (resolved contacts) bukan raw IDs.

```

client.on("message", async (msg) => {
  if (msg.from.endsWith("@g.us")) {
    const botNumber = client.info.wid.user;
    const mentions = await msg.getMentions();
    const diTag = mentions.some(c => c.id.user === botNumber);

    let diReply = false;
    if (!diTag && msg.hasQuotedMsg) {
      const quoted = await msg.getQuotedMessage();
      diReply = quoted.fromMe;
    }
    if (!diTag && !diReply) return;
  }
  // ... proses pesan
});

```

Code Highlight: Streaming Response (SSE)

Efek typing seperti ChatGPT — text muncul token-per-token. Backend pakai **Server-Sent Events**, frontend pakai ReadableStream.

```

const res = await fetch(`${API_URL}/chat/stream`, { /* ... */ });
const reader = res.body.getReader();
const decoder = new TextDecoder();

while (true) {
  const { done, value } = await reader.read();
  if (done) break;
  const chunk = decoder.decode(value);
  for (const line of chunk.split("\n")) {
    if (line.startsWith("data: ")) {
      const { text } = JSON.parse(line.slice(6));
      fullReply += text;
      bubble.innerHTML = renderMarkdown(fullReply);
    }
  }
}

```

3. Brain Tumor Classification

Repo	JerrelAdriel/brain-tumor-classification
Tech Stack	Python Flask, TensorFlow, VGG16 CNN
Akurasi Model	95%+
Origin	Thesis S1 — Universitas Sriwijaya

Apa yang Dilakukan

Aplikasi klasifikasi **tumor otak dari gambar MRI** menggunakan deep learning. Model VGG16 dengan transfer learning, akurasi >95%. Ini dari penelitian thesis S1 di Universitas Sriwijaya (IPK 3.55).

Updates Major

Before	After
UI Bootstrap default, look kuno	Dark theme modern dengan cyan accent
Form upload simple <input type='file'>	Drag & drop zone dengan file preview
Landing page minimal	Hero + stats cards + features grid
Hasil text biasa	Color-coded badge (red/green)

Code Highlight: Drag & Drop Upload

```
[ 'dragover', 'dragenter' ].forEach( ev =>
  dropZone.addEventListener( ev, e => {
    e.preventDefault();
    dropZone.classList.add( 'dragover' );    // highlight saat hover
  })
);

dropZone.addEventListener( 'drop', e => {
  const f = e.dataTransfer.files[0];
  if ( f ) {
    fileInput.files = e.dataTransfer.files;
    showFile( f );    // preview + enable submit
  }
} );
```

UX Flow

- User drag file → drop zone highlight cyan

- Drop → file preview muncul (thumbnail + nama + size)
- Button "Analisis Sekarang" enabled
- Submit → POST ke /predict Flask endpoint
- Hasil ditampilkan dengan badge color-coded

4. Lelang Binar (Second Hand)

Repo	JerrelAdriel/lelang-binar
Tech Stack	React.js, React Bootstrap, Node.js, JWT
Origin	Final Project Binar Academy (2022)
Repo Type	Monorepo (frontend + backend)

Bugs yang Diperbaiki

Bug 1: CardComponent crash saat belum login

jwt-decode throw error kalau input null.

```
// BEFORE - crash
const token = localStorage.getItem('token');
const decode = jwt(token); // crash kalau token null

// AFTER - safe
const token = localStorage.getItem('token');
let decode = null;
try { if (token) decode = jwt(token); } catch (e) { /* ignore */ }
```

Bug 2: Halaman gak bisa di-scroll

```
/* BEFORE - overflow blocked semua scroll */
body { overflow: hidden; }

/* AFTER - horizontal hidden saja */
body { overflow-x: hidden; overflow-y: auto; }
```

Bug 3: p { height: 20px } leak ke semua <p>

Root cause: CSS Modules hanya scope class selector, bukan element selector. Aturan p { ... } di file *.module.css apapun, akan leak ke seluruh app.

```
/* BEFORE - leak global */
p { height: 20px; }

/* AFTER - scoped ke parent class */
.button p { height: 20px; }
```

■ **Note:** Ini bug yang menyebabkan text overlap di success modal nego. Setelah fix, semua <p> di app render proper dengan natural height.

Fitur Baru: Dummy Data Fallback

Backend hosted di Vercel free tier sering sleep. Solusi: **auto fallback ke dummy data**. File src/dummyProducts.js berisi 12 produk lengkap dengan nama, harga IDR, kategori, deskripsi panjang, profile penjual, 1-3 gambar produk dari Unsplash.

```
const getProducts = async () => {
  try {
    const res = await axios.get(url, { timeout: 3000 });
    if (res?.data?.data?.product?.length) {
      setItems(res.data.data.product);
    } else {
      setItems(DUMMY_PRODUCTS);
    }
  } catch (error) {
    setItems(DUMMY_PRODUCTS); // backend down -> dummy
  }
};
```

UI Refactor: Card Grid

Cards di-refactor pakai Bootstrap Row + Col dengan g-3 gap → uniform size, centered, responsive (xs=2 → sm=3 → md=4 → lg=5 columns). Hover lift effect, image aspect-ratio 1:1 cover, price warna merah.

5. Stockop / jStock — Gudang Teknik

Repo	JerrelAdriel/jstockop-teknik
Tech Stack	Next.js 14, React, TypeScript, Tailwind, NextUI
Origin	Internship Magenta MSIB di PT Pelindo (2023-2024)
Architecture	Monorepo: client (Next.js) + server

Apa Itu?

Sistem manajemen stok untuk **Gudang Teknik Pelabuhan Belawan** (PT Pelindo Multi Terminal). Multi-user role (admin/user), CRUD barang, sistem peminjaman & pengambilan barang.

Updates

1. Login Page — Responsive + Modern

Before	After
Fixed grid-cols-2 (tidak responsive)	Mobile-first: grid-cols-1 lg:grid-cols-2
Inline class numbers (ml-40, w-9/12)	Responsive padding & font sizes
Standard form	Modern gradient bg, hero image overlay

2. Demo Mode Login (tanpa backend)

```
const handleLogin = async (event) => {
  event.preventDefault();
  if ((username === 'admin' && password === 'admin123') ||
      (username === 'user' && password === 'user123')) {
    localStorage.setItem('token', 'dummy-admin-token-preview');
    router.push(username === 'admin'
      ? '/pages/admin/dashboard'
      : '/pages/users/home');
    return;
  }
  // ... try real API
};
```

3. Dummy Data 12 Peminjaman + 11 Pengambilan

File `client/app/dummyData.ts` berisi data realistis: Bor Listrik Bosch, Welding Machine Lakoni, Tang Kombinasi Tekiro, Helm Safety MSA, dst — dengan info peminjam, spek alat, jumlah, lokasi proyek, waktu, status.

```
const handleDataLoan = async () => {
  const token = localStorage.getItem('token');
  if (isDummyToken(token)) {
    setDataLoan(DUMMY_LOANS);
    return;
  }
  try {
    const response = await axios.get(API_URL, { headers: { ... } });
    // ...
  } catch (error) {
    setDataLoan(DUMMY_LOANS); // fallback
  }
};
```

4. UI Beautification

Before	After
Sidebar putih, area utama biru, banyak ruang kosong	Gradient slate sidebar, tabel di card putih
NextUI Badge overflow ke main content	Custom inline Counter — clean placement
Tidak ada stats overview	3 stats cards: Total/In Progress/Selesai

6. Jump Game

Repo JerrelAdriel.github.io (in main repo)

Tech Stack HTML5, CSS3, Vanilla JavaScript

Type Browser game (no framework)

Updates Major

Before	After
Single jump pakai CSS animation	Physics-based jumping (gravity + velocity)
Tidak ada double jump	Double jump (2x sebelum mendarat)
Rintangan statis terus-menerus	Random: 65% darat, 35% udara (need double jump)
Game over alert() popup	Proper game over screen dengan skor
Tidak ada skor	Score + high score di localStorage
Tidak responsive	Responsive + touch support mobile

Code Highlight: Physics-based Jump

```
const GRAVITY = 0.7;
const JUMP_VELOCITY = 11.5;
const DOUBLE_JUMP_VELOCITY = 10;
const MAX_HEIGHT = 160;

let karakterY = 0;
let velocityY = 0;
let jumpsLeft = 2;

function jump() {
  if (!isPlaying || jumpsLeft <= 0) return;
  velocityY = (jumpsLeft === 2) ? JUMP_VELOCITY : DOUBLE_JUMP_VELOCITY;
  jumpsLeft--;
}

// Inside game loop (60fps via requestAnimationFrame)
function gameLoop(now) {
  if (karakterY > 0 || velocityY > 0) {
    velocityY -= GRAVITY * dt;
    karakterY += velocityY * dt;
    if (karakterY <= 0) {
      karakterY = 0;
      jumpsLeft = 2; // reset saat mendarat
    }
  }
}
```

```
}  
karakter.style.transform = `translateY(${ -karakterY}px)`;  
}
```

■ **Note:** Kenapa physics > CSS animation? Smooth karena per-frame calculation, collision detection lebih akurat, bisa interrupt mid-air untuk double jump.

Code Highlight: Random Obstacles

```
function spawnObstacle() {  
  // Hindari spawn terlalu dekat  
  if (obstacles.length > 0) {  
    const last = obstacles[obstacles.length - 1];  
    if (last.x > game.offsetWidth - 220) return;  
  }  
  
  const type = Math.random() < 0.65 ? 'ground' : 'air';  
  const el = document.createElement('div');  
  el.className = 'rintangan ' + type;  
  game.appendChild(el);  
  
  obstacles.push({  
    el, type, x: game.offsetWidth + 50,  
    hitTop: type === 'air' ? 80 : 0, // air = di tinggi single jump  
    hitHeight: type === 'air' ? 50 : 55,  
    width: type === 'air' ? 50 : 55,  
  });  
}
```

Rintangan **air** spawn di y=80 (tinggi single jump) → memaksa pemain pakai double jump untuk lewatinya.

7. Infrastructure & Security

GitHub Repositories

Portfolio	github.com/JerrelAdriel/JerrelAdriel.github.io
Agent-JR	github.com/JerrelAdriel/agent-jr
Brain Tumor	github.com/JerrelAdriel/brain-tumor-classification
Lelang Binar	github.com/JerrelAdriel/lelang-binar
Stockop	github.com/JerrelAdriel/jstockop-teknik

Security Cleanup

Token `ghp_V7Jpy...` awalnya tertulis di chat (terexpose). Cleanup yang dilakukan:

- Revoke PAT lama di `github.com/settings/tokens`
- Generate token baru "Claude MCP Jerrel v2" dengan scope sama (`repo`, `read:org`, `read:user`)
- Update token di 2 tempat: `.mcp.json` + Windows User Env Var
- Bersihkan remote URL semua repo (sebelumnya embed token dalam URL)
- Setup `gh` CLI sebagai git credential helper

Benefit Setup Baru

- Token **tidak pernah ke disk lagi** untuk git operations
- **Multi-laptop friendly** — tinggal `gh auth login` di laptop lain
- **Easier rotation** — revoke + buat baru via web UI, no need hunt down `.git/config`

Summary & Lessons Learned

Statistics

Project diupdate	5 + portfolio = 6
Bug fixed	10+
File baru dibuat	25+

Total commits	15+
Languages	JavaScript, TypeScript, Python, HTML, CSS
Frameworks	React, Next.js, FastAPI, Flask, Electron, Bootstrap, NextUI, Tailwind
Tools	Puppeteer, rembg, gh CLI, GitHub MCP

Lessons Learned

1. CSS Modules tidak scope element selectors

Hindari `p { ... }, div { ... }, * { ... }` di file `*.module.css`. Selalu wrap dalam class parent supaya tidak leak ke seluruh app.

2. Fallback patterns untuk demo

Untuk aplikasi yang depend ke external API/backend, selalu siapkan dummy data fallback. Special token (`dummy-admin-token-preview`) untuk distinguish demo vs production mode.

3. Physics-based animation > CSS keyframes

Untuk game-like interaction: lebih kompleks tapi smoother (60fps), collision detection lebih akurat, bisa interrupt mid-animation (e.g. double jump).

4. Multi-platform AI deployment

Satu backend serving multiple frontend (web, WhatsApp, desktop) lebih maintainable daripada 3 codebase terpisah. SSE streaming bikin UX terasa fast.

5. Security best practice

Pakai credential helper (gh CLI / SSH key) supaya token tidak pernah ter-leak ke `.git/config`, terminal history, atau backup.